	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence 3** **Statistical models and research planning** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Name: _____ **Date:** _____

Answer on separate paper and show your reasoning. Use the supplied normal and Student's t reference sheets. Report Student's t critical values and normal critical values from the infinity row to **3 decimal places**, and normal probabilities to **4 decimal places**. For Student's t cutoffs between printed entries, give probability bounds without interpolation. Each criterion is worth **6 points** (**18 points** total); at least **4 points per criterion** are required to pass that criterion.

Questions 1–6 concern independent random observations from normal populations. All critical values and probability questions refer to the sample mean standardized relative to the population mean; critical values have no units.

1. Export inspection costs [C8-2]

An exporter studies average inspection costs using 12 shipments. The population standard deviation is unknown and estimated from the sample.

(a) Determine the degrees of freedom and the positive critical value that leaves 1 percent of the probability to its right. **[1]**

(b) Determine the negative critical value below which 5 percent of the probability lies. Justify its sign and the table column used. **[1]**

2. Market-stall electricity use [C8-2]

A study investigates average electricity use across 25 market stalls. The population standard deviation is unknown and estimated from the sample.

(a) Determine the degrees of freedom. For a central region containing 98 percent of the probability, determine the combined probability outside it and the probability in each tail. **[1]**

(b) Determine the 2 critical values, symmetric about 0, enclosing this central region. Identify the table column used. **[1]**

3. Cold-storage energy costs [C8-2]

An analyst studies average energy costs using 15 cold-storage facilities. The population standard deviation is unknown and estimated from the sample.

(a) Determine the degrees of freedom and give bounds for the probability that the standardized sample mean exceeds positive 2.40. **[1]**

(b) Give bounds for the probability that the standardized sample mean is below negative 2.40, and for the probability that it lies between negative 2.40 and positive 2.40. **[1]**

4. Bottling-line measurements [C9-2]

A bottling plant studies average bottle volume using 16 bottles. The population standard deviation is known to be 3 millilitres.

(a) Choose and justify the distribution model for the standardized sample mean. Determine the negative critical value below which 2.5 percent of the probability lies. **[1]**

(b) Calculate the probability that the standardized sample mean lies between negative 1.00 and positive 1.50. **[1]**

5. Repair-time averages [C9-2]

A workshop studies average repair time using 10 repairs. The population standard deviation is unknown; the sample standard deviation is 4 minutes.

(a) Choose and justify the distribution model, determine the degrees of freedom, and find the positive critical value that the standardized sample mean exceeds with probability 2.5 percent. **[1]**

(b) Estimate the probability that the standardized sample mean lies between negative 1.833 and positive 2.821. **[1]**

6. Comparing parcel-mass studies [C9-2]

2 studies investigate average parcel mass. Study A uses 12 parcels and a known population standard deviation of 0.4 kilograms. Study B uses 12 parcels; its population standard deviation is unknown and estimated by a sample standard deviation of 0.4 kilograms.

(a) Justify the distribution model for each study's standardized sample mean. Determine each model's 2 critical values, symmetric about 0, enclosing the central 95 percent of the probability. Calculate the difference between the positive critical values and explain it in terms of tail behaviour. **[1]**

(b) Study B increases its sample to 61 parcels, with population standard deviation still unknown. State its

model and degrees of freedom, and determine its new critical values enclosing the central 95 percent of the probability. Explain the trend toward the normal model and why sample size alone does not determine the model choice. **[1]**

7. Choose your research option [C10-6]

During the exam: select **1** of the following options and write a very brief description (1–2 sentences) of real-life data that could plausibly follow your selected distribution. **[6]**

(a) A continuous variable modelled by a linear probability density over a stated range.

(b) Waiting times or intervals between events modelled by an exponential distribution.

(c) Normally distributed measurements.

No datasets, sources, calculations, parameter estimates, or detailed model justification are required during the exam.

Online report submission: your PDF report will focus on the **1** selected option and must be submitted virtually through **Open**. This online submission is separate from the brief response written during the exam.